

PHYS400
SPECIAL PROBLEMS IN PHYSICS
PROJECT PROPOSAL

Project Title	Quantifying Information in Gravitational Wave Observations: An Empirical Test of the Flanagan-Hughes Framework
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Abstract

Gravitational wave observations provide information about astrophysical sources, including the masses, spins, and distances of the compact objects. Obtaining this information through Bayesian parameter estimation has become one of the fundamental tasks of modern gravitational-wave astronomy. However, a fundamental question remains unanswered: How much information do these observations actually extract about source parameters, and how well do these empirical results align with existing analytical estimates? Theoretical frameworks such as the Flanagan-Hughes framework estimate this information content by assuming that the information extractable under Gaussian posteriors is logarithmically proportional to the square of the matched filter's signal-to-noise ratio. Despite the merits of these analytical approaches, they have not been systematically tested against the posterior distributions produced by modern parameter estimation processes applied to actual detections. This project addresses this gap by calculating the Kullback-Leibler divergence from the prior distribution to the posterior distribution for the source parameters of binary black hole events in the Gravitational Wave Transient Catalogs. Using publicly available samples from the Gravitational Wave Open Science Center, empirical estimates of information content will be directly compared with analytical predictions to assess the validity of the posterior assumption based on the Gaussian distribution. The results are expected to show that the empirically calculated information content falls short of the analytical estimates, and that these differences are mainly due to non-compliance with the Gaussian distribution and correlations between parameters. Consequently, quantifying this empirical information gap will provide a realistic, data-driven foundation for evaluating the data extraction efficiency of current workflows and the scientific return of next-generation gravitational wave observatories.

Introduction

Gravitational waves are ripples in spacetime caused by the acceleration of massive objects, as predicted by Einstein's general theory of relativity, which he proposed in 1916. The direct detection of these waves has opened a fundamentally new observational window into the universe, enabling us to make measurements of astrophysical sources that cannot

be reached via electromagnetic radiation. The first confirmed detection, GW150914, was made in 2015 by the LIGO-Virgo collaboration and originated from the merger of two stellar-mass black holes (Abbott et al., 2016). This discovery confirmed decades of theoretical predictions and revealed the existence of binary black hole systems in nature.

Since this first detection, the LIGO-Virgo-KAGRA network has conducted four observing runs, producing a growing series of gravitational wave transient catalogs. The most recent release, GWTC-4.0, brings the total number of confident compact binary merger candidates to 219 (Abbott et al., 2016, 2019, 2023, 2024, 2025). Each detected event carries encoded within it physical information about the source, including its component masses, spins, distance, and sky location. The systematic measurement and analysis of this data has become one of the fundamental tasks of modern gravitational-wave astronomy.

The most common tool for extracting source parameters from a detected signal is Bayesian parameter estimation. An observed signal can be used to produce a posterior probability distribution. This is done via Bayesian inference over the source parameters. The process combines prior knowledge with the likelihood of the data under a chosen waveform model. Dedicated samplers such as those implemented in LALInference and its successors are applied to each GWTC event so that these posteriors can be produced (Veitch et al., 2015). The resulting posterior samples, which have been made publicly available through the Gravitational Wave Open Science Center, are the primary data product from which astrophysical conclusions are drawn. With next-generation detectors such as Einstein Telescope and Cosmic Explorer on the horizon, understanding the information-theoretic limits of gravitational wave observations becomes increasingly important, both for characterizing current detections and for evaluating the scientific return of future instruments. Therefore, quantifying the exact amount of information extracted from these observations and systematically comparing it with analytical estimates has become a critical necessity.

This question is naturally addressed within the framework of classical information theory. Shannon (1948) showed that the reduction in uncertainty produced by an observation can be quantified rigorously. In a Bayesian context, the Kullback-Leibler (KL) divergence from the posterior to the prior measures this reduction directly, in units of bits when the logarithm is taken in base two. The analytical framework for this scale was developed by Flanagan and Hughes (1998), who showed that the information extractable about source parameters scales logarithmically with the squared matched-filter signal-to-noise ratio. Their framework is elegant and compact, but it rests on the assumption that posterior distributions are approximately Gaussian, which need not hold for real signals analyzed with realistic waveform models and prior choices.

Despite the usefulness of this framework, it has not been systematically tested against actual GWTC posterior distributions. We still don't know if the Gaussian assumption is true in real life, how the information changes with different signal-to-noise ratios and

mass configurations, and how close real detections are to the Flanagan-Hughes analytical approach. Addressing these issues requires the combination of the theoretical framework and the posterior samples now available across the GWTC catalogues. It is this gap that the present project aims to fill.

Research Question

This project addresses a specific question that has not yet been systematically tackled in the literature: Across all gravitational wave events detected to date, how many bits of information have been extracted about the source parameters in total?

While Flanagan and Hughes (1998) established a theoretical framework for estimating this quantity using the Kullback-Leibler divergence under a Gaussian posterior approximation, no study has yet applied this framework directly to the posterior samples available in the GWTC catalogs.

We hypothesize that the actual empirical information extracted will be close to, but systematically below, the analytical Flanagan-Hughes predictions. This shortfall is expected to grow for parameters that are strongly correlated or poorly constrained. To comprehensively explore this, the project will also address the following sub-questions: (1) What is the physical origin of the discrepancies between empirical data and theoretical predictions? (2) What do these measurements imply about the information extraction efficiency of current and next-generation observatories?

Methodology

The theoretical foundation of this project is the framework developed by Flanagan and Hughes (1998). The exact expression for the information extractable about source parameters is given by the integral:

$$I_{\text{source}} = \int d\theta p(\theta|s) \log_2 \left(\frac{p(\theta|s)}{p(\theta|p(s))} \right) \quad (1)$$

where $p(\theta|s)$ is the posterior distribution given the observed signal, and $p(\theta|p(s))$ is the prior distribution conditioned solely on the detector noise power spectral density. This expression represents the Kullback-Leibler (KL) divergence from the posterior to the prior in units of bits. Computing this integral directly from real data is the central goal of this project.

Flanagan and Hughes (1998) introduced an analytic estimate by assuming that the noise is Gaussian and stationary, and that the posterior is well approximated by a Gaussian.

Under these conditions, the information integral reduces to:

$$I_{\text{source}} \approx \frac{N_{\text{param}}}{2} \log_2 \left(1 + \frac{\rho^2}{N_{\text{param}}} \right) \quad (2)$$

where ρ is the matched-filter SNR and $N_{\text{param}} = 15$ for a binary black hole system, covering the component masses, spin magnitudes and orientations, and extrinsic parameters. This approximation holds well in the high-SNR regime but becomes less reliable at low SNR.

The data are drawn from the gravitational wave transient catalogs (Abbott et al., 2019, 2023, 2024) via the Gravitational Wave Open Science Center (Abbott et al., 2021). Parameter estimation data files for GWTC-3 events are accessed through the corresponding data release (LIGO Scientific Collaboration et al., 2023) and read using the `pesummary` library (Hoy and Raymond, 2021). Each file contains posterior and prior samples for over 100 parameters. However, instead of calculating the divergence for every variable that is available, the project’s focus is on selecting and analysing a subset of parameters that are considered to be physically connected and relevant to the source dynamics. Furthermore, the `C01:IMRPhenomXPHM` waveform approximant is used consistently across all events. This specific choice is strictly necessary because valid prior samples—which are required to compute the KL divergence—are consistently available under this approximant. As an initial validation step, the analysis will first be calculated and tested for individual parameters based on a single event. It will then be expanded to a broader set of GWTC events.

The empirical information content is computed by numerically evaluating the KL divergence for the selected source parameters separately. The posterior and prior samples are fitted with a one-dimensional Gaussian Kernel Density Estimator (KDE) using the `scipy` library. However, a critical limitation addressed in this project is that summing these one-dimensional marginal KL divergences may not strictly yield the true total multidimensional information due to inter-parameter correlations. The extent to which this marginal sum approximates the total extracted information remains an open question that this study will explicitly explore.

To ensure the reliability of the KDE-based estimates, the results will be cross-validated against independent methods, such as a Monte Carlo estimator and a discrete entropy calculation. The empirical estimates will then be compared against the analytical predictions from Equation 2. Where discrepancies arise, their origin will be investigated in terms of the breakdown of the Gaussian approximation, inter-parameter correlations, or prior choices. All computational and statistical tasks will be implemented in Python.

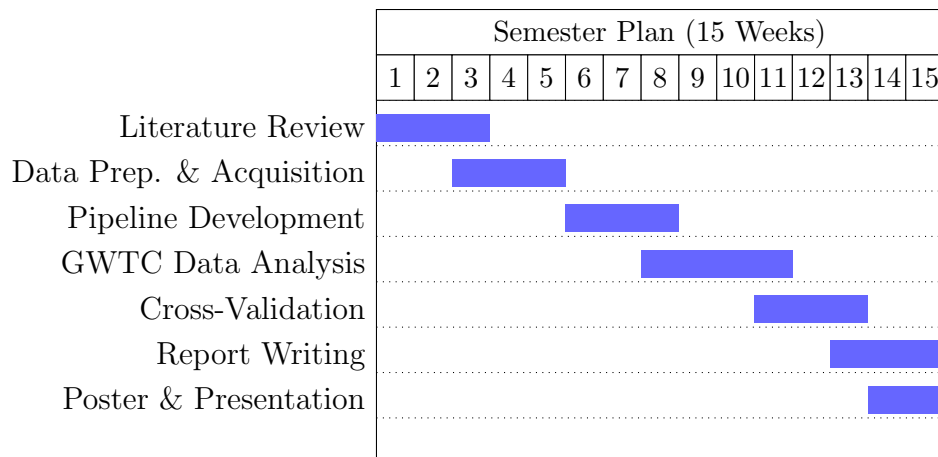
Expected Results

Based on the theoretical framework and the analytical approximation given in Equation 2, we expect ground-based interferometers, operating with signal-to-noise ratios (SNR) ranging from 5 to 100, to yield information contents of $10 \lesssim I_{\text{source}} \lesssim 120$ bits per event. In contrast, the future space-based LISA detector, with anticipated SNR values between 10^3 and 10^5 , is expected to achieve $200 \lesssim I_{\text{source}} \lesssim 400$ bits. However, we hypothesize that the empirical information computed from real GWTC data will fall systematically below these analytical Flanagan-Hughes upper bounds. This shortfall is anticipated to be particularly pronounced for events with low SNR and highly non-Gaussian posterior distributions.

The actual empirical results may differ significantly from the theoretical predictions because the analytical formulation relies on several idealizing assumptions. Specifically, it assumes that the detector noise is perfectly Gaussian and stationary, that waveform models depend linearly on the source parameters near their true values, and that the resulting posteriors are strictly Gaussian. In reality, these assumptions frequently break down for complex astrophysical signals. Furthermore, our approach of summing one-dimensional marginal KL divergences neglects inter-parameter correlations, meaning the resulting sum may over- or under-estimate the true multidimensional information content.

Despite these uncertainties, measuring this empirical information gap will provide a significant contribution to the field. Ultimately, this study will show exactly how much useful information our current analysis methods can pull from real gravitational wave signals. By measuring this accurately, we can set a realistic, data-driven baseline to help us understand the true scientific potential of future observatories like the Einstein Telescope and Cosmic Explorer.

Timeline



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